AARON HORVITZ

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Portfolio: GitHub / Stack Overflow

Security Clearance: Previous Public Trust/ Suitability (IRS), US Citizen

GEN AI | APPLIED STATISTICS | MACHINE LEARNING | DATA SCIENCE

I am a Senior Data Scientist and AI Engineer with a background in applied statistics, currently building production-grade GenAI systems at PwC. During my four years as a GS-14 Data Scientist within the IRS's RAAS division, I engineered deterministic models and automated audit pipelines.

Statistical Analysis: Time Series (SARIMAX), Classification, Multivariate Analysis, Regression, Forecasting
Data Engineering: ETL Scripting, Pipeline Automation
Programming & Databases: Python, (R, SAS, & JMP for Academic Work), Linux
Tooling: VS Code, JupyterLab, DBeaver, GitHub/GitLab
AI Coding Tools: Codex, Claude Code, Gemini Code Assist
Deep Learning & ML: Scikit-Learn, Statsmodels, Pandas, NumPy, PyTorch

GEN AI & DATA SCIENCE EXPERIENCE

Gen AI Python Systems Engineer | 10/06/2025 – Present
(PwC) Pricewaterhouse Coopers in Advisory Services | Houston, TX

<u>Enterprise & Agentic AI Deployment:</u> As part of PwC's AI Factory and Agent OS team, I assist the team with building out and deploying production-grade Generative AI systems.
<u>Object-Oriented Pipeline Architecture:</u> Developed modular Python classes that seamlessly integrated into the team's overarching data ingestion pipeline, operating within a hybrid infrastructure to manage cross-platform transformation workloads and consolidate components into a unified system.
<u>ETL & Complex Data Ingestion</u>: Engineered a robust ETL process to parse and validate multi-sheet service order and renewal workbooks generated by the client's sales representatives. Implemented extensive business logic and strict data verification rules using Pydantic to handle highly variable, human-entered data.
<u>Software Engineering & CI/CD:</u> Wrote verifiable, production-grade Python code adhering to enterprise software engineering standards. Authored comprehensive unit tests and participated in peer code reviews.

Statistician (Data Scientist) | GS-14, Step 4 **| 11/08/2021 – 09/30/2025**
Internal Revenue Service (IRS) in (RAAS) Research, Analytics, and Applied Statistics | Houston, TX

<u>LLM Deployment Support</u>: Supported internal implementation of LLM-driven chatbot interfaces, providing IRS employees with streamlined access to over a million pages of the Internal Revenue Manual using RAG and other techniques.
<u>Email Classification & LLM Analytics</u>: Led the creation, project design, and coding of various email classification models, integrating Large Language Models (LLMs) and classical machine learning techniques, streamlining operational efficiency by effectively categorizing taxpayer inquiries for clean vehicle energy tax credits.
<u>Technical Leadership & Executive Briefings:</u> Pitched and secured unanimous executive sponsorship from IRS RAAS Directors to architect and lead a new automation initiative. Conducted multiple high-level presentations to federal executives, successfully translating complex data engineering concepts into operational enforcement capabilities.
<u>Audit Analysis Application (Python / ETL)</u>: Led the development of an automated financial data processing application for Revenue Agents. Engineered the underlying Python ETL pipeline to support analytical processing and structured data validation, extracting complex financial data from various structured and unstructured sources.
<u>Time-Series Forecasting & Trend Analysis</u>: Developed forecasting models (SARIMAX, LOWESS) to predict the volume of taxpayer interactions with a web-based Virtual Assistant. Created a novel in-time parameter tuning solution.

Data Scientist | 08/05/2019 – 11/05/2021
McDermott International, Ltd | Houston, TX

Workforce Planning & Survival Analysis: Pair-coded a Workforce Management-Planning model with my boss to help management visualize when employees would be mobilizing and demobilizing from the company's project locations, yards, fabrication facilities, and offices located globally. As part of this project, I built an automated ETL process that scraped submitted monthly operational data, performed data cleaning and transformation, and loaded it into the data warehouse.
Time Series Forecasting: Created McDermott's weekly COVID-19 forecast and long-term scenarios for 12 countries to monitor impacts on multi-billion-dollar active projects. Wrote an automated Python script (utilizing Pandas, Scikit-learn, and Statsmodels) to extract and process raw epidemiological data directly from the Johns Hopkins GitHub repository. Engineered the predictive generation using a custom-coded auto-ARIMA and various statistical learning methods to provide accurate alternatives to early IHME models. Built a process that automatically executed the weekly run and output the results directly into a Word document and presentation format for executive distribution.
Developed Internal Python Statistical Toolset: Created a standard statistical toolset in Python for the team that displayed the summary of fit, shrinkage tools, parameter estimates, diagnostics (actual vs. predicted, studentized residuals vs. Cook's distance, histogram of the residuals), leverage, and marginal model plots for regression modeling. Used StatsModels and Scikit-learn.
Conducted Graduate-Level Statistics & Coding Training: Conducted a weekly class in graduate-level applied statistics and Python coding. Topics included advanced multivariate regression, leverage points, shrinkage, etc. Modeled the Python training after MIT's EdX course "MITx-6.00.1x: Introduction to Computer Science and Programming Using Python." This effort resulted in each participating team member earning the MITx-6.00.1x certificate.

Data Scientist | 11/18/2018 – 07/29/2019
Avangard Innovative (Purchased by Waste Management) | Houston, TX

Computer Vision: Coded, trained, validated, and tested Convolutional Neural Networks (CNN's) for an image classification model to identify recycled materials.
Real-Time Predictive Modeling: Built a statistical model to predict waste compactor capacity based on discrete, high-noise pressure readings taken during ram cycles. Engineered the model to fit a smooth trend that automatically adjusted in real-time as new sensor data was ingested.
API Integration & Normalization: Deployed a model to estimate waste container fullness by normalizing continuous pressure readings pulled via API and converting them into actionable percentage-full metrics for client optimization.
Statistical Model: Created a statistical model (an ANOVA with a Tukey Range Test) to evaluate relative machine performance using StatsModels in Python.
Python Script and Forecasting: Analyzed inventory flow after building SQL queries into raw JD Edwards data periodically copied into the data warehouse. We used this data to build a time series model to forecast inventory. The script also provides real-time inventory reporting.

EDUCATION

MS Analytics/Data Science | Texas A&M University (Department of Statistics) | **Completed in 2018**
BS Economics | University of Houston | **Completed in 2010**
BS Political Science | Texas A&M University | **Completed in 2000**
Houston Community College | Various Mathematics Courses

DeepLearning.AI	**Completed 2026**	PyTorch for Deep Learning Professional Certificate
MITx 6.00.2x	**Completed 2020**	Introduction to Computational Thinking and Data Science
MITx 6.00.1x	**Completed 2017**	Introduction to Computer Science and Programming Using Python

SAS Certified Associate: Programming Fundamentals Using SAS 9.4: Link to Cert on Credly

ADDITIONAL PROFESSIONAL & MILITARY EXPERIENCE

Recruiting Agency Founder - The Confidential Advisor LLC | Feb 2010 - Nov 2018 | Houston, TX
Executive Recruiter & Business Development - Qualitec Technical Services | Apr 2005 – Jun 2008 | Houston, TX

United States Army | Sep 2000 - Nov 2004 | **Honorable Discharge** | **Good Conduct Medal**
US Army Airborne Ranger (2nd Ranger Battalion) & Infantry Soldier (1-23 Stryker Brigade)

TECHNICAL PROJECTS

Agent Gatsby (US Treasury's Challenge) | *https://github.com/AaronNHorvitz/Agent-Gatsby*

Agent Gatsby is a local-first AI pipeline built for US Treasury's supplemental Great Gatsby challenge. Engineered a deterministic state machine using open-weight models to orchestrate a multi-stage workflow: evidence extraction, exact-substring quote verification to prevent hallucination, and bounded expansion loops for strict word-count compliance, culminating in Spanish and Mandarin translations.

adaptive-lasso (Statistical Computing) | *https://github.com/AaronNHorvitz/AdaptiveLASSO*

Built a Python package for adaptive LASSO in linear regression with a Statsmodels-style workflow, including packaging, tests, documentation, and a demo notebook. The project translates an established statistical method into a cleaner, reproducible implementation for applied modeling, teaching, and experimentation.

LexiChess (Local-First AI Chess Arena) | *https://github.com/AaronNHorvitz/LexiChess*

LexiChess is a chess tournament and evaluation platform for benchmarking LLMs, chess agents, and custom engines. Engineered a reproducible match pipeline with deterministic legal-move validation, SQLite-backed game storage, tournament scheduling, Elo-style rating updates, PGN exports, Stockfish analysis, and pluggable providers for local models. Includes a FastAPI-powered live broadcast interface with showmatch controls, moderation workflows, referee commentary, and interactive human-vs-agent play.

Labels On Tap (Local-First AI Verification) | *https://github.com/AaronNHorvitz/Labels-On-Tap*

Developed a secure, local-first ML compliance engine for TTB alcohol label verification. Combined deterministic string matching (RapidFuzz) with a custom docTR/DistilRoBERTa ensemble and visual typography preflighting to automate regulatory triage. Containerized for AWS deployment with FastAPI, Docker, and a durable batch-processing queue for human-in-the-loop review.

TopicMiner (Text Classification Tool) | *https://github.com/AaronNHorvitz/TopicMiner*

Built a modular Python toolkit for high-volume deterministic email classification using classical NLP – LDA, Doc2Vec + K-Means, and Word2Vec + K-Means for topic discovery, with Random Forest and XGBoost classifiers on top of a shared base-class interface. Designed as a deterministic, auditable alternative to LLM-based classification on narrow-taxonomy tasks, where fine-tuned classical methods consistently outperform LLM zero-shot at a fraction of the inference cost and latency.

iSignify (Bioinformatics & AI Pipeline) | *https://github.com/AaronNHorvitz/iSignify*

Integrated Gemma 3 for on-device interpretation of complex bio-statistical results, ensuring data privacy for biosecurity applications. Implemented a custom k-mer comparison algorithm in Python to identify unique genomic signatures across massive datasets without pre-computed databases, optimizing for computational efficiency.